

SensBlue Monarch — Datasheet

Cellular IoT controller with LTE-M / NB-IoT and
GNSS

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1. General description

The ATRONIA SensBlue Monarch is a ruggedised, low-power IoT controller board designed for cellular-connected remote sensing, monitoring and control. It pairs an Espressif ESP32-WROOM-32E microcontroller with a Quectel BG95-M3 modem, giving the device native Wi-Fi 802.11 b/g/n, Bluetooth 4.2 BR/EDR + BLE, LTE-M Cat.M1, NB-IoT Cat.NB1/NB2, EGPRS fallback and a multi-constellation GNSS receiver in a single compact package.

The board integrates the peripherals typically needed by field devices: an on-board BME280 environmental sensor, a 256-byte AT24C02 EEPROM, an OLED display header (SSD1306 128×32 by default), two opto-coupled digital inputs, two high-side digital outputs, two analog inputs software-selectable between 0-10 V and 4-20 mA current-loop modes, and a half-duplex RS485 transceiver with 15 kV ESD protection.

Power is flexible by design. An LM66200 ideal-diode input manager auto-selects between USB, a Li-ion battery, an external DC input and a solar-panel input, feeding a synchronous buck regulator (LMR33630) for the main 3.3 V rail. A dedicated Li-ion charger (BQ25176) manages battery charging from any available source, and an on-board boost converter (LMR62421) generates the 12 V rail used by the 4-20 mA current loop and available as an external output.

The SensBlue Monarch is fully programmable through the Arduino IDE via the standard Espressif ESP32 core and a companion library (SensBlueMonarch) that exposes every on-board peripheral through an idiomatic API. Hardware documentation and firmware sources are open under the MIT licence.

2. Key features

Processing and wireless

- Espressif ESP32-WROOM-32E-N8 module: Xtensa dual-core LX6 @ 240 MHz, 520 KB SRAM, 8 MB flash
- Wi-Fi 802.11 b/g/n (2.4 GHz), station and soft-AP modes
- Bluetooth 4.2 BR/EDR and Bluetooth Low Energy

Cellular and satellite positioning

- Quectel BG95-M3: LTE-M Cat.M1, NB-IoT Cat.NB1/NB2, EGPRS 850/900/1800/1900 MHz fallback
- Multi-constellation GNSS: GPS, GLONASS, BeiDou, Galileo
- External antenna via U.FL (single antenna covers LTE and GNSS)
- Nano-SIM socket, 1.8 V / 3 V compliant

Sensing and storage

- Bosch BME280 — temperature (± 0.5 °C), relative humidity (± 3 %), barometric pressure (± 1 hPa)
- AT24C02 EEPROM — 256 bytes, I²C @ 0x50
- SSD1306 OLED display header (I²C, 128×32 default; 128×64 supported by software)

Industrial I/O

- 2 × opto-coupled digital inputs (EL357N), inputs rated up to 30 V, threshold ~3 V

- 2 × high-side digital outputs (BCX56 + NPN drivers), sourcing V_EXT to load
- 2 × analog inputs, software-selectable 0-10 V or 4-20 mA (shared configuration)
- RS485 half-duplex (ISL83483), up to 250 kbps, 15 kV HBM ESD protection

Power management

- Multi-source input: USB (5 V), Li-ion battery (2.5-4.4 V), external DC (up to 5.5 V), solar panel (up to 6.5 V)
- LM66200 ideal-diode input auto-select, low-drop, reverse-current blocking
- BQ25176 Li-ion charger with power-path management
- LMR33630 synchronous buck regulator — main system rail
- LMR62421 boost converter — 12 V, feeds 4-20 mA loop and available externally
- On-board voltage monitors: battery, panel, USB, and switched output

Programming and software

- USB-UART programming via Silicon Labs CP2102N (auto-reset / auto-boot)
- Arduino IDE support via Espressif ESP32 core
- SensBlueMonarch companion library (open source, MIT licence)
- 12 reference examples covering every on-board peripheral

3. Applications

The SensBlue Monarch targets deployments where cellular connectivity, low power, and industrial-grade I/O are simultaneously required. Representative applications:

- Cellular remote monitoring — agriculture, water metering, environmental telemetry
- Smart cities — waste, parking, noise, air-quality gateways
- Perimeter security — solar farms, industrial fencing, remote assets
- Off-grid, solar-powered field devices with intermittent connectivity
- Asset and fleet tracking using integrated GNSS and LTE-M
- Industrial signal conditioning — 4-20 mA / 0-10 V front-ends to cloud
- Predictive maintenance — vibration, temperature, pressure logging

4. Block diagram

The following block diagram summarises the main functional groups and their interconnections. The ESP32 sits at the centre, communicating with the BG95 modem over a dedicated UART, with the on-board peripherals through a shared I²C bus, and with the industrial I/O through a PCAL6416A GPIO expander and dedicated ADC pins.

USB (5 V) CP2102N	Battery (Li-ion) BQ25176 charger	External DC LM66200 ideal-diode	Solar panel (3.3 W, 6 V)
LMR33630 buck → 3.3 V system rail · LMR62421 boost → 12 V (4-20 mA loop / OUT)			
ESP32-WROOM-32E-N8 Wi-Fi + BLE 4.2 240 MHz · 8 MB flash		Quectel BG95-M3 LTE-M / NB-IoT / EGPRS + GNSS multi-constellation	
BME280 T / RH / P	PCAL6416A I ² C GPIO expander	AT24C02 EEPROM 256 B	SSD1306 OLED header (128×32)
2 × DIG_IN opto-isolated EL357N	2 × DIG_OUT high-side BCX56 + NPN	2 × Analog IN 0-10 V / 4-20 mA mux (TS5A23157)	RS485 ISL83483 half-duplex 15 kV ESD
Nano-SIM socket · U.FL antenna · Piggyback SPI/I ² C header (30-pin Hirose) · RGB LED · 2 × button			

Figure 2. SensBlue Monarch functional block diagram

5. Pin and connector descriptions

The SensBlue Monarch exposes a mix of pluggable screw-terminal blocks for field I/O and dedicated headers for expansion and programming. All connectors are labelled on the silkscreen with the reference designators in this section.

5.1 Screw terminal blocks (I/O)

Eight 2-position 3.5 mm pluggable terminal blocks (XY302V-3.5-2P) and one 3-position block (XY302V-3.5-3P) provide the field-side interface for power, isolated I/O, analog inputs and RS485.

Designator	Signals	Description
H9	V_EXT, GND	External DC input (up to 5.5 V), used as an alternative to USB / battery
H10	V_PANEL, GND	Solar panel input (up to 6.5 V Voc)
H11	V_BAT, GND	Li-ion battery pack terminals
H12	DIG_IN_1, GND	Isolated digital input 1 (opto-coupled, up to 30 V)
H13	DIG_IN_2, GND	Isolated digital input 2 (opto-coupled, up to 30 V)
H14	DIG_OUT_1, V_EXT	High-side digital output 1 (switches V_EXT to load)
H15	DIG_OUT_2, V_EXT	High-side digital output 2 (switches V_EXT to load)
H18	AIN_1, AIN_2	Two analog inputs (shared 0-10 V / 4-20 mA mode selection)
H20	RS485-A, RS485-B, GND	RS485 differential pair with common reference

5.2 Programming and status

Designator	Type	Description
CN2	USB-C 6-pin	USB device port (5 V input + data), programming and Serial console
CN3	2-pin	Reset / boot header (advanced use, normally not populated by end-user)
SW2	Push-button	User button (BTN) — mapped to ESP32 GPIO4
SW3	Push-button	Reset button (RESET)
LED	RGB 5050	Status LED — driven via PCAL6416A expander bits 13-15

5.3 Display and antenna

Designator	Type	Description
H1	U.FL / IPEX	External antenna (LTE and GNSS multiplexed inside the BG95)
H3	Nano-SIM socket	SIM card holder (push-push, hinged)
H17/H19/H21	FFC 7-pin	I ² C OLED display interface (128×32 SSD1306 supported by default)

5.4 Expansion header — H4

H4 is a 30-pin, 0.4 mm-pitch Hirose DF40 board-to-board connector that exposes a piggyback expansion bus. Two additional GPIOs are routed through the PCAL6416A expander (P1_0, P1_1) and can be used for board-select or interrupt lines.

Signal	Direction	Notes
3.3V, GND	power	Board 3.3 V rail; 3.3 V total budget shared with on-board peripherals
SPI-MOSI / MISO / SCK	ESP32 → piggyback	GPIO13 / GPIO12 / GPIO14 — piggyback-only SPI
SPI-SS	ESP32 → piggyback	GPIO5 (strapping-safe idle-high)
I ² C SDA / SCL	shared	GPIO21 / GPIO22 — same bus as on-board peripherals
UART2-TX / RX	muxed	GPIO17 / GPIO16 — shared with RS485 via USART_SEL bit
IOEXP P1_0 / P1_1	via expander	General-purpose piggyback GPIOs (open-drain via I ² C)
RESET (piggyback)	ESP32 → piggyback	Optional reset signal to daughter-card
INT (piggyback)	piggyback → ESP32	Optional interrupt line (GPIO19)

6. Absolute maximum ratings

Stress ratings above those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only; functional operation of the device at these or any other conditions above those indicated in the operational sections is not implied. Exposure to absolute-maximum conditions for extended periods may affect device reliability.

Parameter	Symbol	Min	Max	Unit
USB input voltage	V_USB	-0.3	6.0	V
External DC input	V_EXT	-0.3	6.0	V
Solar panel input (Voc)	V_PANEL	-0.3	7.0	V
Battery pack voltage	V_BAT	-0.3	4.4	V
12 V boost output current	I_12V	—	200	mA
Digital input voltage (isolated)	V_DIN	-0.3	30	V
Digital output source current (per channel)	I_DOUT	—	500	mA
Analog input, 0-10 V mode	V_AIN	-0.3	10.5	V
Analog input, 4-20 mA mode	I_AIN	—	24	mA
RS485 differential voltage	V_RS	-7	+12	V
Storage temperature	T_STG	-40	+85	°C
Junction temperature	T_J	—	+125	°C

7. Recommended operating conditions

Parameter	Min	Typ	Max	Unit	Notes
USB input voltage	4.5	5.0	5.25	V	USB spec compliant
External DC input	3.6	5.0	5.5	V	via LM66200 ideal diode
Solar panel (Voc)	—	6.0	6.5	V	AK135×165, 3.3 W polycrystalline recommended
Li-ion battery pack	3.0	3.7	4.2	V	Single cell, thermistor-monitored charge
Operating temperature	-25	+25	+75	°C	Limited by BG95-M3 module
Ambient humidity	—	—	95	%RH	Non-condensing
I ² C bus speed	—	100	400	kHz	Some external OLEDs limited to

Parameter	Min	Typ	Max	Unit	Notes
					100 kHz

8. Power consumption

The following figures are derived from constituent component datasheets and are indicative only. Values marked "TBD by testing" require validation on production hardware; final numbers will replace them in a future revision of this datasheet.

Operating mode	Typical current @ 3.7 V	Notes
Deep sleep (ESP32 hibernate + BG95 PSM + all rails but essential off)	~50 μ A (TBD)	Wake sources: BTN, RTC timer, DIG_IN
Idle, LTE-M attached (ESP32 modem sleep, BG95 registered)	~10 mA (TBD)	Periodic tracking-area updates
Active, radios off	~40 mA	ESP32 @ 240 MHz, all peripherals on
Active + Wi-Fi TX	~250 mA avg	Peaks to ~500 mA during TX bursts
LTE-M TX @ +23 dBm	~240 mA avg	Peak ~500 mA, budget for battery brown-out
GNSS cold acquisition	~75 mA (TBD)	Assist-GPS reduces time to first fix
OLED refresh (128 $\times$ 32, all pixels)	+15 mA	During display update; base current 5 mA

Peak currents during LTE transmit bursts exceed 500 mA. The Li-ion charge path and the LMR33630 buck are sized for this; battery packs paired with the board must be able to source at least 1 A at 3.6 V (i.e. cells with >1 C burst discharge rating).

9. Peripheral interfaces

9.1 Wi-Fi and Bluetooth

Wi-Fi and Bluetooth are provided by the ESP32-WROOM-32E module through an integrated PCB antenna. Wi-Fi supports 802.11 b/g/n in the 2.4 GHz ISM band, station and soft-AP modes, WPA/WPA2/WPA3 security, and up to 150 Mbps physical-layer rate. Bluetooth supports Bluetooth 4.2 BR/EDR and Bluetooth Low Energy (BLE), with concurrent Wi-Fi and BLE operation possible but subject to airtime sharing.

9.2 Cellular — Quectel BG95-M3

The BG95-M3 supports multi-mode LTE Cat.M1, Cat.NB1, Cat.NB2 and EGPRS with worldwide band coverage (LTE B1/B2/B3/B4/B5/B8/B12/B13/B18/B19/B20/B25/B26/B27/B28/B66/B71/B85). The modem is powered through the VBAT_2 rail (gated by the on-board expander) and controlled by ESP32 UART1 (115 200 baud default). Power-on and reset sequences use the PWRKEY pin driven by the PCAL6416A expander. The modem embeds a TCP / UDP / HTTP(S) / MQTT / SSL / DTLS stack accessible over AT commands.

9.3 GNSS

The BG95 embeds a dedicated multi-constellation GNSS receiver supporting GPS L1 C/A, GLONASS, BeiDou B1I and Galileo E1. The GNSS output shares the LTE antenna via an internal front-end. NMEA-0183 sentences are available on the modem's GNSS UART, routed to ESP32 UART2 through the on-board USART_SEL multiplexer (bit 12 of the expander). To activate the GNSS path, GPS_EN (expander bit 0) is asserted and USART_SEL is set to 1.

9.4 Environmental sensor — BME280

The Bosch BME280 provides calibrated readings of ambient temperature, relative humidity and barometric pressure over the I²C bus at address 0x77. Typical accuracy is ± 0.5 °C, ± 3 % RH and ± 1 hPa. The sensor draws less than 4 μ A in low-power weather-monitoring mode (forced measurement at 1 Hz) and less than 0.1 μ A in sleep. The library exposes readTemperature(), readHumidity() and readPressure() methods that abstract the mode-forcing sequence.

9.5 Isolated digital I/O

Two isolated digital inputs use EL357N opto-couplers to provide up to 30 V input range and 2.5 kV isolation between the field side and the microcontroller. When the input is energised the ESP32 pin is pulled low; the library inverts this logic, so readDigitalInput() returns true when the input is active.

Two digital outputs are implemented as high-side switches driven by BCX56 medium-power transistors and NPN pre-drivers. The output switches the V_EXT supply to the load; the load is connected between DIG_OUT_x and its return (GND on the field side). Continuous current up to 500 mA per channel is supported provided the V_EXT source can supply it.

9.6 Analog inputs

Two analog inputs share a common current/voltage front-end selected by INPUT_IV_SEL (expander bit 11). In 0-10 V mode a resistor divider (10 k Ω / 2.49 k Ω) scales the input to the ESP32 ADC range; in 4-20 mA mode a 100 Ω shunt converts the loop current to a voltage sampled by the ADC. The +12 V boost is enabled automatically when 4-20 mA mode is selected in software, with a 1 s settle delay before the first valid reading. Both channels share

the mode selection — mixed modes on the two channels simultaneously are not supported by the hardware.

Parameter	0-10 V mode	4-20 mA mode	Notes
Input range	0 to 10 V	0 to 24 mA (limited by shunt)	Full scale software-selectable
Input impedance	>100 k Ω	100 Ω shunt	0-10 V input divider dominant
ADC resolution	12 bit	12 bit	ESP32 SAR ADC, ~11 ENOB typical
ADC linearity	TBD	TBD	Non-linear near rails; calibrate for production
Loop compliance	N/A	12 V nominal	From on-board boost

9.7 RS485

An ISL83483 3.3 V half-duplex RS485 transceiver connects to ESP32 UART2 via a bus multiplexer (TS5A23157) shared with the GNSS path. The driver enable (DE/!RE combined) is on GPIO18. The transceiver is rated for 250 kbps signalling with slew-rate control for RFI reduction, ± 15 kV HBM ESD on the bus pins, and true fail-safe receiver behaviour. Fail-safe biasing is provided by pull-up/pull-down resistors on the on-board differential lines.

9.8 I²C peripheral bus

A single I²C bus (GPIO21/22) hosts the on-board PCAL6416A expander (0x20), BME280 (0x77), AT24C02 EEPROM (0x50), the OLED display header (0x3C), and any I²C peripherals attached to the piggyback header. Default bus speed in the SensBlueMonarch library is 100 kHz for maximum compatibility with third-party OLED modules and cabled peripherals; the ESP32 supports up to 400 kHz Fast-mode and 1 MHz Fast-mode Plus subject to signal integrity.

9.9 USB-UART

USB access is provided by a Silicon Labs CP2102N USB-to-UART bridge, connected to ESP32 UART0. The DTR and RTS lines are wired through the standard ESP32 auto-reset / auto-boot network (Q3 and Q13), so ordinary Arduino-CLI or esptool.py flashing works without manual intervention. The default USB VID/PID is Silicon Labs (10C4:EA60); the CP2102N internal EEPROM can be reprogrammed for a custom VID/PID if required.

10. Power management

10.1 Input sources and diode-OR

Four independent input sources feed the SensBlue Monarch: USB (5 V, from CN2), a Li-ion battery pack (H11), an external DC input (H9, 3.6-5.5 V), and a solar panel (H10, up to 6.5 V open-circuit). An LM66200 ideal-diode input auto-selector picks the highest available voltage source with negligible voltage drop and blocks reverse currents between sources. This allows the board to operate seamlessly across the loss and restoration of any input.

10.2 Battery charging

A Texas Instruments BQ25176 single-cell Li-ion charger manages battery charging from any active input source. The charger integrates power-path management: the system is always supplied from the highest priority input, and only surplus current is used to charge the battery. This lets the device continue to operate at full power even with a fully depleted battery, provided one of the wired inputs is present.

Default charging termination is set for a 4.2 V / cell Li-ion chemistry. The charge current is configured by an on-board resistor. Battery voltage is monitored on the ESP32 through a divider on GPIO34, gated by VBAT_2_EN so the divider does not draw current when the battery is not being sampled.

10.3 Main regulator

The 3.3 V system rail is generated by an LMR33630 synchronous buck converter, rated for 3 A output current and up to 36 V input. The regulator's high efficiency (>90 % typical at load) and wide input range accommodate all supported input sources without pre-regulation. A TLV803 voltage supervisor holds the ESP32 in reset until VCC is stable.

10.4 12 V boost — 4-20 mA loop and external output

An LMR62421 boost converter generates the 12 V rail used to power the 4-20 mA current loop, and available at pin H4 as a switched output for external use. The boost is gated by BOOST_EN (expander bit 3) and needs approximately 1 s to reach regulation after enable. The library enableBoost() (aliased as enable12V()) waits this time internally. Output current is limited to approximately 200 mA.

10.5 Auxiliary rails

Rail	Purpose	Enable	Notes
VCC	Main 3.3 V system rail	always on	Powers ESP32, expander, CP2102N
VCC_2	Modem support rail	expander bit 6 (VCC_2_EN)	Level shifters and pull-ups
VCC_3	On-board peripherals rail	ESP32 GPIO2 (VCC_3_EN)	BME280, EEPROM, OLED — 1 s settle
VBAT_2	BG95 modem power rail	expander bit 4 (VBAT_2_EN)	Cut when modem is off to save power
12 V	4-20 mA loop / OUT	expander bit 3	LMR62421 boost, 1 s settle

Rail	Purpose	Enable	Notes
		(BOOST_EN)	time

11. Mechanical and environmental

11.1 Board layout



Figure 3. SensBlue Monarch — top view

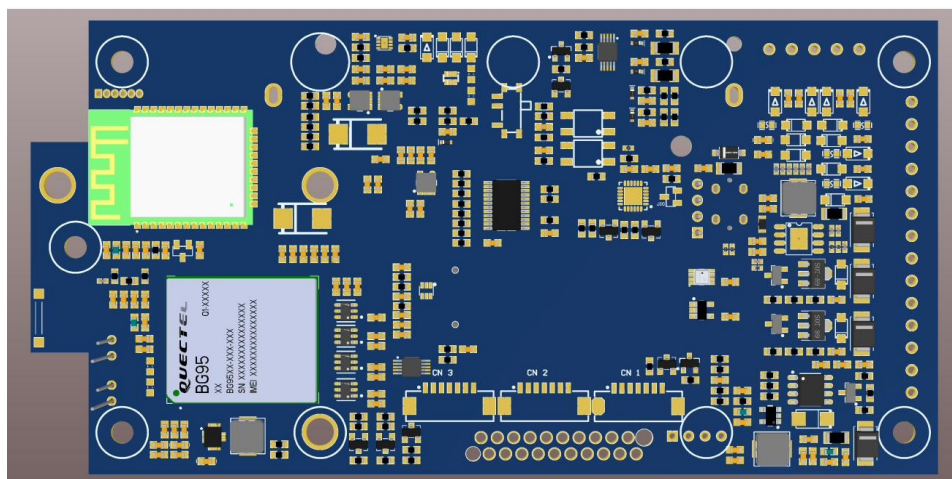


Figure 4. SensBlue Monarch — bottom view

11.2 Dimensions and mounting

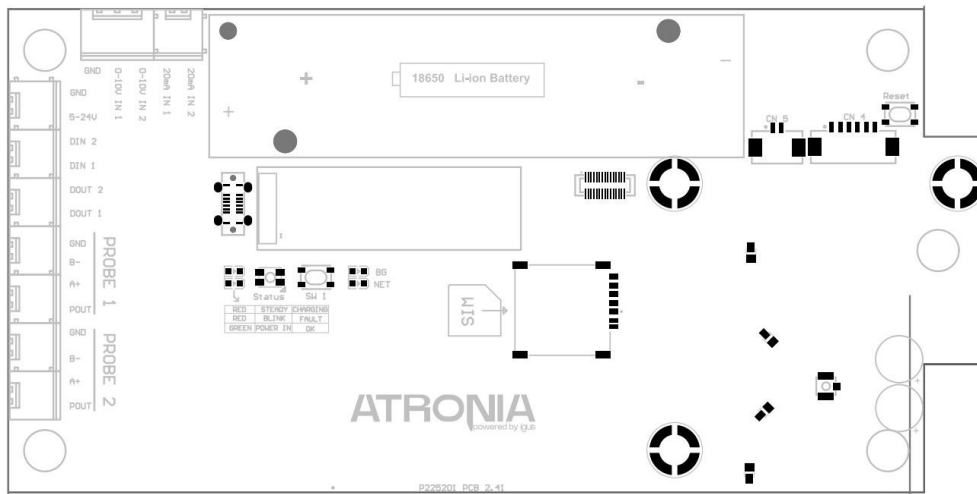


Figure 5. Board dimensions and mounting hole locations (drawing scale 1:1 at 100 dpi)

Four mounting holes (ECT818009999 press-fit brass inserts) accept M3 screws. Mounting hole coordinates and board outline are defined in the accompanying mechanical drawing (see repository docs/schematics/).

11.3 Environmental ratings

Parameter	Value	Notes
Operating temperature	-25 °C to +75 °C	Limited by BG95 module
Storage temperature	-40 °C to +85 °C	
Operating humidity	0 to 95 %RH	Non-condensing
Vibration	10 Hz - 500 Hz, 2 g	TBD by testing
Ingress protection	IP20 (bare board)	Enclosure required for outdoor deployment
Altitude	up to 2 000 m	Standard atmospheric conditions

11.4 Enclosure compatibility

The SensBlue Monarch is mechanically designed to fit the Gainta G258 series of polycarbonate enclosures (160 × 80 × 55 mm). The board's mounting-hole pattern aligns with the internal PCB standoffs of the enclosure, so the board slides in without extra hardware.

Enclosure model	Ingress rating	Lid	Notes
Gainta G258MF	IP65	Opaque	Baseline sealed enclosure, wall-mount

Enclosure model	Ingress rating	Lid	Notes
			flanges (M4)
Gainta G258CMF	IP65	Transparent	OLED readable through the lid; matches ATRONIA solar-panel kit
Gainta G258MF-IP67	IP67	Opaque	Higher ingress protection variant
Gainta P258MF-IP68	IP68	Opaque	Fully sealed variant for immersion-rated deployments

12. Software support

The SensBlue Monarch is fully supported from the Arduino IDE through the standard Espressif ESP32 core plus the ATRONIA-published SensBlueMonarch companion library. The library abstracts every on-board peripheral through an idiomatic API and includes twelve reference examples covering LEDs, buttons, isolated digital I/O, analog inputs (with mode switching), RS485, the BG95 modem, GNSS, BME280 environmental readings, the OLED display, EEPROM storage, and battery state-of-charge estimation.

Both the library and its examples are released under the MIT licence. Contributions and issues are welcomed at the project's GitHub repository.

Resource	Location
Companion library	https://github.com/sensblue-monarch/SensBlueMonarch
Board schematics	docs/schematics/ in the library repository (11 sheets, JPG)
Pinout reference	docs/pinout.md in the library repository
Arduino ESP32 core	https://github.com/espressif/arduino-esp32
Wiki and tutorials	https://wiki.atronia.eu/SENSBLUE_MONARCH
Product enquiries	suporte@atronia.eu

13. Accessories and ecosystem

The following accessories are available from ATRONIA to complete a field-ready deployment of the SensBlue Monarch. Each item is designed and validated for direct integration with the board and the Gainta G258 enclosure family.

13.1 Enclosures (Gainta G258 series)

- Gainta G258MF (IP65, opaque lid) — baseline sealed enclosure with flanges for wall mounting
- Gainta G258CMF (IP65, transparent lid) — allows OLED readability without opening the box
- Gainta G258MF-IP67 (higher ingress protection)

13.2 Solar panel with G258 direct-fit mount

A 3.3 W / 6 V polycrystalline solar panel supplied pre-assembled onto a Gainta G258 transparent-lid enclosure. The panel bonds to the transparent lid and connects to the board's V_PANEL screw terminal through a pre-fitted sealed cable gland. This is the recommended off-grid power option for outdoor deployments.

13.3 Antennas

Accessory	Description
Adhesive PCB antenna, U.FL	Low-profile flexible sticker antenna covering the LTE and GNSS bands used by the BG95-M3; U.FL termination, adhesive backing for mounting on the inner wall of the enclosure
High-gain N-type antenna	External antenna for demanding coverage environments; N-type connector for outdoor mounting
U.FL to N-type pigtail	Bulkhead pigtail that connects the board U.FL to an external N-type antenna through a sealed panel gland

13.4 Battery

For deployments exposed to sub-zero temperatures, ATRONIA supplies low-temperature 18650 Li-ion cells rated for discharge down to -40 °C and charging down to -20 °C, matched to the on-board BQ25176 charging profile. The 18650 form factor mounts directly to the ATRONIA-supplied battery holder that fits inside the G258 enclosure.

14. Ordering information

Item	Value
Product code	TBD
Internal project	P225201
Hardware revision	2.41

Item	Value
Standard supply	Board only. Antennas, enclosure, solar panel and battery available as separate accessories (see section 13)
Compatible enclosure	Gainta G258 series (160 × 80 × 55 mm, IP65 / IP67 / IP68)
Regulatory status	In progress — contact ATRONIA before deploying in regulated markets
Minimum order qty	Contact ATRONIA

15. Revision history

Document rev	Date	Changes
0.1 (preliminary)	September 2026	Initial preliminary release. Includes accessories section (Gainta G258 compatibility, solar panel, antennas, battery). Values marked TBD to be finalised by testing on production hardware.

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